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▲TEXAS
Jonny Durand in Texas in 2016 – on a trip where he hoped to break the 800km mark. The HG world record stands at 764km, flown in Texas on 3 July 2012
Photo: Michael Clark

WEATHER

THE INVISIBLE RAILWAY

Jörg Korner has compared long-distance flying in Brazil and Texas – with some remarkable results. By Ed Ewing

Jörg Korner has been a pioneering data scientist for decades. By training he approaches everything he does by applying scientific principles first, often taking long-held beliefs or assumptions before working out what's really happening. Over his career he has applied this to helping start-ups for the Swiss government through to exploring biological sciences.

He's also a paraglider pilot and owns a garage-workshop with a good Wi-Fi connection, which means he can spend long hours diving into the abundance of digital flying data that we, as a community, have created over the last 25 years. His latest project? Exploring the world's longest paraglider flights, combining this data with historic weather data and using it to reveal more about those rare record days in both Brazil and Texas than we've ever known before.

Brazil vs Texas

If you want to fly an open-distance hang gliding or paragliding world record then your main choice is Brazil or Texas.

On paper, Texas wins. Both the current hang gliding and paragliding world records were set there. The hang gliding world record of 764km (475 miles) was set from Zapata, Texas on 3 July 2012 by Dustin Martin, with Jonny Durand flying a mere 3km less. The paragliding world record of 613km (381 miles) was set by Sebastien Kayrouz on 20 June 2021.

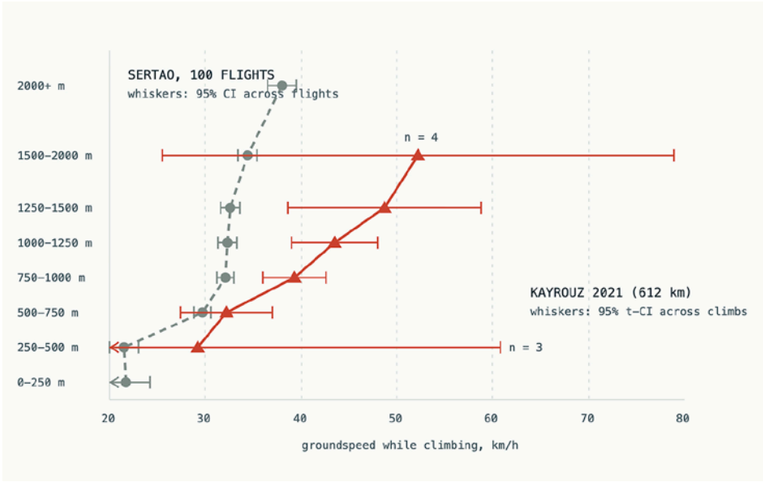
But the logistics of flying in Texas have always been difficult. You need to tow, you need a big group of pilots, there are a lot of Badlands with locked gates and potentially difficult landowners with gun licences. Pilot tales of hanging out in middle-of-nowhere places like Zapata for weeks at a time waiting for "the day" also put pilots off. Traditionally it has been an "expedition" destination.

And indeed that is why Texas weather guru Gary Osoba, a legendary pioneer hang glider pilot and now sailplane pilot in the USA, established what was called the World Record Encampment back in the early 2000s. He invited pilots to come and fly from Zapata, Texas in mid-summer and offered to sort the towing logistics. The lure was the Texas Dryline, a weather phenomenon which, on the right day, can provide a racetrack of lift stretching nearly 1,000km across Texas.

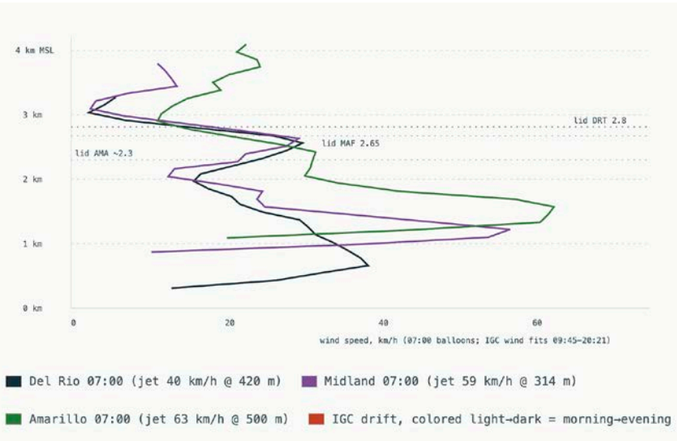
Over the next few years lots of very good pilots did hear the call, including hang gliding legend Manfred Ruhmer, who flew 700.8km for a new world record on 17 July 2001. And Will Gadd, the Canadian pilot who broke the paragliding world record twice towing from Zapata; his furthest was 423km on 23 June 2002. A decade later Dustin and Jonny flew their never-bettered 764km world record. Two years after that Ozone took a team of paraglider pilots there, and came home with six 400km flights but no world record.

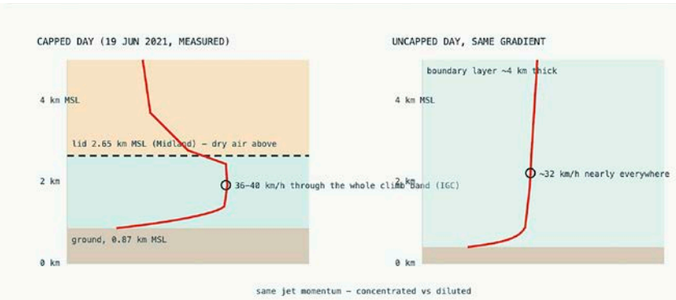
However, despite those big flights, and given it is potentially such an amazing location, Texas attracts surprisingly little activity. At least that is when you compare it to Brazil. In the mid- to late 2000s most paraglider pilots who wanted to have a crack at setting a new world record headed to the northeast of Brazil. XC dragon-slayers flocked to Quixadá where they could launch from a granite morro (monolith) in impossibly strong wind at 7am, soar for an hour until the thermals started, then ping over the back, frisbeeing across the sertão, the Brazilian outback, until landing at sunset.

The logistics were easier, the weather in late-September/October offered a consistent wind in the right direction, and there was little to no airspace to worry about. If you got bored, you could head to the coast and go kitesurfing.



▲GROUNDSPEED WHILE CLIMBING
Jörg Korner compared 100 NE Brazil tracklogs with world record holder Seb Kayrouz's tracklogs in Texas. After a lot of work he identified the one clear parameter that sets Texas apart: the average speed while climbing. In Texas it is 41.5km/h, versus 34.5km/h in Brazil
Illustration: Jörg Korner















▲THE RAILWAY
Top: This shows the morning wind trace from three different weather balloons in the Texas Dryline area. It clearly shows a 'jet' of wind at 40-60km/h, between 750-1,500m. As the day progresses this jet rises, but its speed stays steady

Bottom left: The measured tracklog from Seb Kayrouz's world record flight. It shows Seb's groundspeed while climbing is 36-40km/h though the whole climb band. The results are the same for Seb's 500km flight in 2020 and his 600km flight in 2021. Because there is not a lot of flying, there are not a lot of tracklogs to work with

Bottom right: This shows the average groundspeed while climbing for 100 400-580km flights flown in Brazil. It is remarkably consistent at 32km/h through the whole climb band

Jonny Durand in flight on his 760k day in 2012

Seb Kayrouz and partner after his world record flight in 2021

From 2007 to 2019 the Brazilians, always flying as a team – "an organism in the air" – took the paragliding world record from Will Gadd's 423km and increased it in stages to 582km in 2019. (Nevil Hulett was the first to fly 500km. He flew it in South Africa in 2008 and held the world record for seven years.)

Seb Kayrouz

At the same time as pilots were going big in Brazil, US pilot Sebastien Kayrouz was scouring the literature to learn about the Texas Dryline and saving up to buy a winch. In 2020 he flew 501km from footlaunch on his EN-D. Flying alone he was the epitome of the lone wolf. A sales exec for a financial software company he'd learnt to fly in Lebanon in 1996 and his previous PB was 232km, flown in Oklahoma the year before. He didn't fly competitions, didn't log his flights regularly on XContest and pilots outside local XC circles in the USA didn't know his name. But he was dedicated and knowledgeable.

A year later he was back. This time he towed up on his new Ozone Enzo 3 (he had bought his winch) and on 20 June 2021 – the longest day of the year – he flew 609.9km straight-line. He also set the 556.8 km straight distance to a declared goal record on the same flight.

On the ground

On a recent flight back from a paragliding trip to Colombia data scientist Jörg Korner had a layover in Dallas, Texas. As he flew in he could see the line of cumulus clouds that form along the Texas Dryline going for miles. Clear air to the west, perfect cloud streets to the right.

Intrigued he looked up Seb Kayrouz's record flight and, taking a rental car spent the day driving to Seb's landing spot. Why, he wondered, is no one flying Texas? Every year, he thought, the Swiss federation send their best pilots to Brazil on an XC camp, and many fly very long distances. But no 600km flights (yet!).

More than that, he wondered, how come one pilot, operating on their own, managed to fly more than 600km, when hundreds of pilots have flown in Brazil, scores of them have flown 500km, but not one has made 600? The numbers, he decided, did not make sense. He sent a message to Seb Kayrouz through one of the platforms, and Seb agreed to meet him that Sunday afternoon, between parenting duties. "We had a long talk and I asked him lots of questions about his flight. Seb had worked out how that air behaves by flying in it – and I realised I had the tools to go after what he was describing." That something was about how the air in Texas behaves – something Jörg came to call the Texas Railway.

Groundspeed/climbing

At home in Switzerland Jörg set about comparing the two world record sites: Brazil versus Texas. He collected100 excellent tracklogs from Brazil, but only had two tracks from Texas – Seb's record flight and 500k flight from the year before.

After a lot of work he identified the one clear parameter that sets Texas apart, and puts it on a different level.

"It is the average groundspeed while climbing," he explained. "In Texas it is 41.5km/h versus 34.5km/h in Brazil."

Jörg looked at 25 different parameters – from final glide to thermal spacing to average climb rate – and when comparing Brazil and Texas they were all more or less similar. Except for this one factor: how fast the pilot covers ground as they circle in lift.

He discovered that in Brazil a pilot's groundspeed while climbing is pretty much the same from start-of-climb to cloudbase; in Texas it accelerates as the pilot climbs: from 43.5km/h at 1,000m to 52km/h at top of climb (which is not cloudbase – but where Seb left his climb).

"Same climb, same glide like a good Sertão day, except that the air he climbed in kept moving faster the higher he went," explained Jörg. "What kind of sky does that to a circling paraglider – and can you know it is coming before you launch?"

That question is what sent Jörg down a rabbit-hole – and he came up with an answer, which might just revitalise long-distance flying in Texas for a new generation of record-hungry XC paraglider pilots.

The Texas Dryline

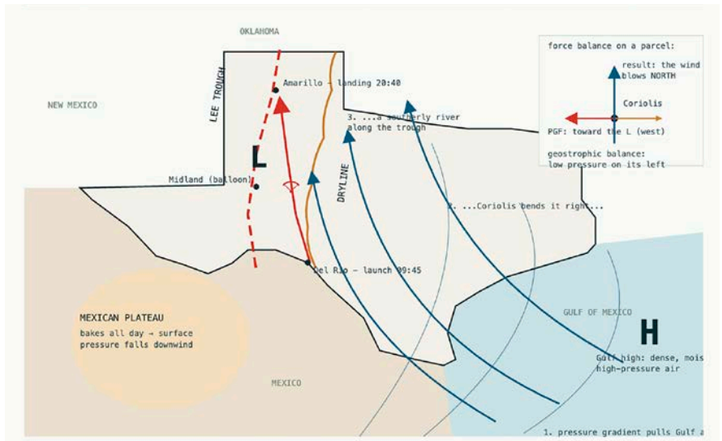
In a long article published on his personal website this month, Jörg explains that through examining weather data combined with available tracklogs he has identified a low-level 'jet' of fast-moving air that exists in the southerly wind that flows

▲CLOCKWISE FROM MAIN
Flying Quixadá in its footlaunching heyday in 2017
Photo: Kiwi Johnston

Bottom left: A classic Brazilian afternoon sky
Photo: Kiwi Johnston

Top left: The Texas sky on Seb Kayrouz's 600k day
Photo: Seb Kayrouz





▲BRAZIL
Sebastian Benz on a 592km flight in the Brazilian sertão on 30 September 2025. Sebastian set a world record for free distance using up to three turnpoints. Analysis of his track shows he benefitted from an early strong wind, an early take-off, and brilliant flying!
Credit: Sebastian Benz

▲TEXAS
The weather system for the creation of record conditions in Texas. A dryline is where dry air (here off the Mexican Plateau) meets moist air (here off the Gulf of Mexico) at the ground – a line on the map. Aloft, the same boundary is the capping inversion – the lid
Credit: Jörg Korner

through Texas under specific conditions. This jet is what Manfred Ruhmer, Dustin and Jonny and Seb Kayrouz naturally took advantage of when they set their world records. The absence of the jet on otherwise good XC days is also what explains why many pilots who have flown Texas have 'only' flown 400km and have left disappointed, wondering what the fuss is about.

A record +600km day in Texas, Jörg posits, start with a typical Texas Dryline day. A dryline is a narrow atmospheric boundary that separates a moist air mass (in this case the Gulf of Mexico to the east) from a dryer air mass (in this case the dry air over the Mexican Plateau).

The dryline season is late April to early July. When a high pressure sits over the Gulf of Mexico southerly winds set up, which blow straight across Texas. "Coriolis bends it right until it flows north along it," explained Jörg, "a southerly tailwind running day and night. The dryline is the surface edge where the moist Gulf air wedges out against the dry continental air."

The result is cloud streets that go on for miles, which in themselves would make for a great XC day. What makes a record day is the low-level jet. This, Jörg suggests, is a wind that builds with height and runs day and night. Importantly, it is capped by an inversion. Without the cap the day would overdevelop – storms are common along the Texas Dryline. It is the inversion that keeps the jet concentrated below it, which in turn allows soaring pilots to hitch a ride.

Finding a record day

Jörg's analysis has led him to create a checklist to identify if a dryline day has 600k potential.

1. The season and pattern. Late April to early July. Look for low pressure over eastern New Mexico–West Texas, high pressure over the Gulf and tightly packed isobars between. Tropical systems east of Texas can strengthen the set-up; an approaching cold front kills it.

2. The river. Check Del Rio, Midland and Amarillo soundings for a strong southerly low-level jet: 160-200°, ideally 40+ km/h a few hundred metres above the ground. No morning jet, no railway.

3. The lid. Look for the inversion where temperature rises and dewpoint falls sharply. Ideally you want a 1.5-2.5km deep working band beneath it: enough depth for usable thermals while keeping the wind concentrated.

4. Will it hold? Calculate the surface temperature needed to break the inversion. If the forecast maximum stays comfortably below it, the lid should hold all day. If temperatures reach it and storms are forecast along the dryline, the set-up is likely to break down.

5. The fuel. You still need enough heating beneath the inversion to generate reliable lift. Around 400J/kg of CAPE below the lid can produce 1.5–3m/s thermals and a flat cumulus field without punching through the cap.

6. Where to start. Find the dryline and aim roughly 50-150km east of it. Farther west means stronger wind and lower ceilings; farther east means higher ceilings and softer conditions. Start well south, around Del Rio–Hill Country latitude, to maximise the flying day and exploit the terrain early and late.

Brazil or Texas?

Jörg admits that flying in Brazil is the most dependable XC arena there is, with steady trade winds, and more 400km+ flights than anywhere. Pilots have flown near-500k here on EN-Bs. It offers an early start, 7am, so the flying window rivals Texas, where pilots cannot start before 9.30am. There are great landings, good logistics and straightforward retrieves. The trade-off is lift lines run wide apart and drift gently (~30km/h), so you fly higher and give up the top-end speed.

Texas meanwhile demands specific conditions: a capped low-level jet that offers the fastest-climbing air on record. And it packs the lift close: the shallow capped layer spaces its cloud streets tightly – roughly 2-8km apart against the sertão's 6-12km – so climbs come often (~4 min apart on the record) and you can stay low and still reach the next one. Mid-summer offers 14-hour days, and towards the end of the flight the spectacular Palo Duro Canyon, second only in size to the Grand Canyon, offers a parting gift: it exhales at dusk, for a broad upwelling of buoyant air for an outsized final glide at sunset.

(On Seb Kayrouz's record flight his final glide at 8pm lasted half an hour and carried him 38km at a glide ratio of 73:1! He still had a 40km/h tailwind as the sun set. According to weather guru Gary Osoba, who has flown over the Palo Duro canyon three times at the end of long sailplane flights, this is not an anomaly – it is a regular feature).

The challenge

Jörg hopes his work will invigorate the quest for paragliding world records. The record books – and the data – show that Texas is the place to go, he says. With his new insight into how the area works, today's weather-forecasting tools and high-performance wings, coupled with some of the serious XC talent out there, a whole new set of world records is waiting to be set. Who, he asks, is willing to try? XC

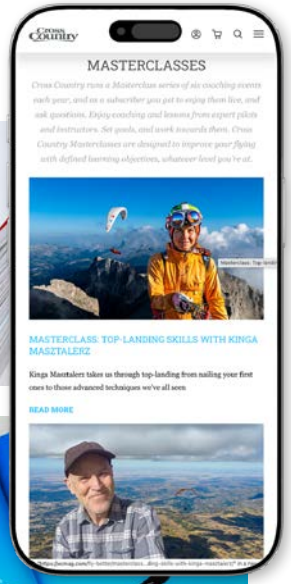
READ MORE

"Turning free public data into flying knowledge, for any arena." You can read Jörg Korner's article, The Invisible Railway, comparing Brazil and Texas, including all data sources, at the link.
joerxc.org/invisible-railway



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